

MERRYLAND HIGH SCHOOL, ENTEBBE

S.2 MATH HOLIDAY WORK

ITEM ONE

In Hoima city there are various secondary schools with different populations. One of the schools is SPAH with a population of 720 students. It's a mixed school with both girls and boys, with a boarding and a day section. If 20% of the students at St Patrick's high school are male.

Task:

- a) What is the percentage of females in the school?
- b) How many more females than males are in the school?
- c) If 25% of the female students are day scholars, how many female students are in boarding?
- d) If all male students are in boarding and 3 students share a decker, how many deckers are needed to accommodate all the male students?

ITEM TWO

Two learners were given a task of plotting the following points on the grid. A (0,4) B (2,2) C (4,2) D (2,0) E (4,-2) F (0,-1) G (-4,-2) H (-2,0) I (-4,2) and J (-2,2). Before they plotted the points, Jane told Musa that when plotting, for point A you move 4 units to the right of the origin and no movement along the y-axis from the origin. For point C you move 2 units to the right from the origin and units parallel to the y-axis in the positive direction. Musa said no for point A there is no movement along x-axis, you only move 4 units along the y-axis. While for point C you move 4 units from the origin on the x-axis, then two units parallel to the y-axis.

Task

- a) Comment with reasons on Jane's explanation of plotting the points.
- b) Using Musa's explanation, plot the coordinates.
- c) Join the points to form a polygon.

ITEM THREE

Your friend owns a rectangular piece of land of which he wants to construct a house with a rectangular cross section leaving a path 2m around the house base. The path occupies an area of 76 m² and he also wants the dimensions of his house floor to differ by 3m.

He also owns a soft drink depot where he sells Crates of soda and boxes of juice. Tom bought 4 crates of

soda and three boxes of juice and paid Shs 110,700. Tom's brother bought 3 crates of soda and 4 boxes of juice and paid Shs 122,400 but he is not contented about the money he paid claiming that he bought almost similar items as Tom but he has paid much money.

Task:

a) (i) Determine the size of the floor base of your friend's house.

(ii) How big is your friend's piece of land?

b) (i) Help Tom's brother to be contented with the payment he made.

(ii) How much would Ann pay if she buys 2 crates of soda and 2 boxes of water from the same depot?

ITEM FOUR

On their Prom party, five students plan to buy their class teacher a gift. The initial plan was that each one buys their own gift to give to the class teacher but later they realized that If they do it as a group would be cheaper So they mobilized themselves and each person contributed as follows;

Student A contributed one hundred thirty -three thousand two hundred fifty shillings.

B contributed $1\frac{1}{2}$ of the amount of A. C contributed 20% more than that of A.

D contributed a fraction which is $\frac{1}{5}$ less that of B of the amount of A contributed.

E. She contributed the balance required to make shs 800,000 which was the cost of the gift . On the day of Prom, the money was given to student E to pick the gift from the nearby shop. From school, she used a boda-boda moving at 10km per hour, they moved for 30 minutes due east and then 15 minutes due north to reach the shop.

Tasks:

a). Help to direct another student to locate the shop from school in coordinate form.

b). What amount did each student contributed and who of the five contributed more money.

d). In which other way could they contribute the money more fairly and how much would each contribute that way?

ITEM FIVE

Two friends formed a partnership and agreed to deal in animal feeds. Ben contributed seven hundred and fifty thousand shillings while Carey contributed UGX.900,000 . They agreed to save $\frac{1}{5}$ of their gross profit in their joint account and further use 20% of the remainder to buy materials for new products and there after, share the net profit in the ratio of their initial contribution. At the end of their first year of operation, they made gross profits amounting to UGX 3,160,000 and Carey deposited Bens share on his account but Ben was not sure if it was the actual amount he was supposed to get so he sought for your help to avoid biasness. Ben sent you to withdraw some money from his phone out had forgotten the pin however he remembered that he had saved it as 179 in decimal base which he converts to base five to get the actual pin.

Task

(a) Show Ben how to determine the actual amount that was to be deposited in his bank account.

(b) How did you manage to get the actual pin in order to make the withdrawal from Ben's phone?